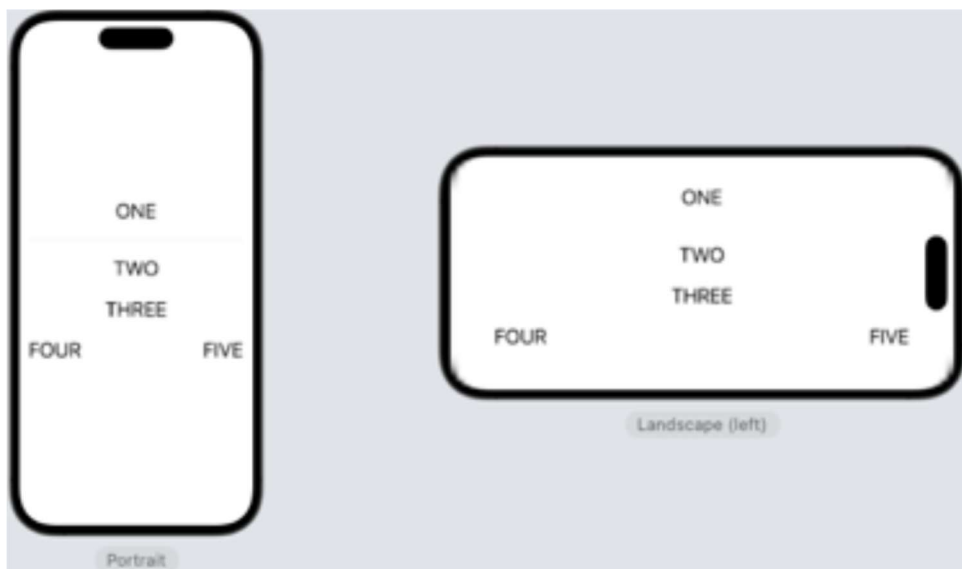


M.C.A. (2nd Semester)**Mobile Application Development using iOS****Question Bank****Unit: 1 Basic SwiftUI Views and Controls**

1. Compare SwiftUI with UIKit framework. Explain any four advantages of SwiftUI.
2. Write Swift code and design sample user interface using ZStack. Draw the preview user interface and explain the solution.
3. Design SwiftUI user interface using Label and Image view. Use at least two different modifiers for Label and Image and explain your code.
4. Create user interface using Toggle and Stepper. Write code snippet, draw diagram and explain your solution.
5. Write SwiftUI code to design following user interface using VStack, HStack, Text, Divider, and Spacer.



6. How Swift prevents common programming errors? Explain any five.
7. Create user interface using Image view with resizable, scaleToFill, frame, clipShape, and opacity modifiers. Write code snippet, draw diagram and explain your solution.
8. Write code snippet and explain how to apply group of styles using ViewModifier.
9. Write Swift code snippet and explain Button and Segmented Control with appropriate example.
10. Explain custom parameter attribute that constructs views from closures using appropriate example and code snippet.

11. Draw the user interface designed by "Code Block-1" and "Code Block-2". Compare and discuss the output.

<pre> var body: some View { // Code Block-1 VStack (spacing: 28) { Text("ONE") Divider() Text("TWO") Text("THREE") HStack { Text("FOUR") Spacer() Text("FIVE") } } .font(.largeTitle) .padding() } </pre>	<pre> var body: some View { // Code Block-2 VStack (spacing: 28) { Text("ONE") Text("TWO") Text("THREE") HStack { Text("FOUR") Spacer() Divider() Text("FIVE") } } .font(.largeTitle) .padding() } </pre>
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12. What is the use of Text view? Design sample user interface and explain Text view with font and frame modifiers.
13. Write Swift code and design sample user interface using VStack and HStack. Draw the preview user interface and explain the solution.
14. Design SwiftUI user interface using Color Picker and Date Picker. Write Swift code and explain the solution.
15. Step-by-step explain, how to integrate UIKit in to SwiftUI.
16. Write Swift code snippet and explain Text Field and Text Editor with appropriate example.
17. With appropriate example, explain how to add SwiftUI to legacy UIKit app.
18. Which view is used to display one or more lines of read-only text? Design sample user interface and explain that view with lineLimit and truncationMode modifiers.

Unit: 2 List Views, Scroll Views, and Advanced Components

1. What is the use of ScrollView? Write code snippet and explain vertical and horizontal scrolling using ScrollView.
2. Write code to display employee information using custom rows in a List. Explain your solution.
3. Write Swift code for searchable Employee list. Also explain your code.

4. When to use adaptive, flexible, and fixed grid items? Discuss with example and Swift code.
5. Analyze the following code snippet and explain ScrollView.

```

ScrollView([.horizontal, .vertical]) {
    VStack(spacing: 20) {
        ForEach(0..<20) { row in
            HStack {
                ForEach(0..<20) { column in
                    Text("\(row), \(column)")
                        .foregroundColor(.white)
                        .frame(width: 100, height: 100)
                        .background(Color.red)
                        .cornerRadius(50)
                }
            }
        }
    }.frame(maxWidth: .infinity)
}.font(.largeTitle)

```

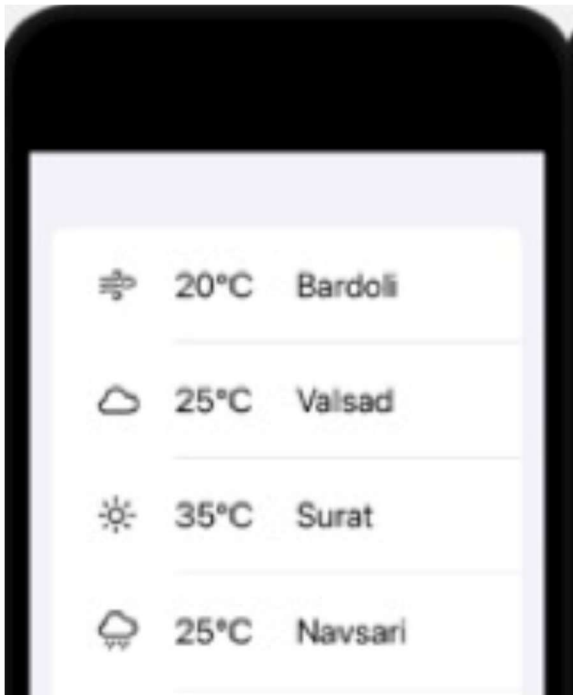
6. With appropriate example and code snippet, explain how to add a row to a List.
7. What is search scope with respect to searchable list? Explain with appropriate example and code snippet.
8. What is programmatic scrolling? When it is useful? Write Swift code and explain ScrollViewReader.
9. What is the use of List view? Compare List and ScrollView. Discuss with appropriate examples.
10. Which functionalities are available in edit mode of a List? Explain editable list with appropriate example and code snippet.
11. Explain use of expanding list and disclosure group with appropriate example and necessary code snippet.
12. Explain editable List with code snippet and appropriate example.
13. What is lazy stack? Compare LazyHStack, LazyVStack with HStack, VStack.
14. Explain one advantage and one limitation of lazy stack.
15. What is SwiftUI widget? How it is useful? Explain three key components of SwiftUI widget.
16. Create editable collection of TodoTask which allows user to update TodoTask. Write Swift code for required type and view. Explain your solution.

17. Which SwiftUI lazy view is used to display tabular content? Give example and explain with necessary code snippet.

18. Analyze the following code snippet and explain it.

```
List {
    ForEach(searchResults, id: \.self) { food in
        LabeledContent(food.name) { Text("\(food.category.rawValue)")
        }
    }
    .searchable(text: $searchText,
                prompt: "Search Test")
    .searchScopes($scope) {
        Text("All").tag(FruitSearchScope.all)
        Text("Fruit").tag(FruitSearchScope.food(.fruit))
        Text("Meat").tag(FruitSearchScope.food(.meat))
        Text("Vegetable").tag(FruitSearchScope.food(.vegetable))
    }
    .navigationTitle("Foods")
}
```

19. Write SwiftUI code to display weather information using List view as given below:



20. Explain how to use custom rows in a List view with necessary code snippet and example.

21. Step-by-step explain, how to create SwiftUI widget for existing app.

Unit: 3 Navigation

1. What is NavigationStack? Considering appropriate example, write Swift code to implement static navigation. Explain your solution.
2. Explain the use of NavigationPath with appropriate example and code snippet.
3. Demonstrate use of NavigationSplitView with necessary code snippet.
4. Explain NavigationStack and NavigationLink with appropriate example and code snippet.
5. Design solution using init(sidebar:detail:) and demonstrate multi-column navigation. Write necessary code snippet and explain.
6. What is TabView? Write code snippet and explain how to use it.
7. When to use Swift enum? Write an example enum which confirms Identifiable, CaseIterable, and Hashable. Also explain how to use that example enum in Swift.
8. With appropriate example and code snippet, explain multi-column navigation with sidebar and detail.
9. Analyze the following code snippet and explain it.

```

struct ContentView: View {
    var body: some View {
        TabView {
            VStack() {Text("Home View")}
            .tabItem {Text("Home view")
                Image(systemName: "house.fill") }

            NavigationStack() {Text("Profile view")
                .navigationTitle("Profile")}
            .tabItem {Label("Profile", systemImage:
                "person.fill") }

            NavigationStack() {Text("About view")}
            .tabItem {
                Text("About view")
                Image(systemName: "info.circle") }
            .badge("12")
        }
    }
}

```

10. Write Swift code snippet and explain navigationDestination(for:destination:) modifier.
11. Design solution using init(sidebar:content:detail:) and demonstrate multicolumn navigation. Write necessary code snippet and explain.
12. With appropriate example and code snippet, explain switching tab programmatically.
13. Write necessary Swift code and justify the following statement:

“Apple provides the NavigationPath struct, as a type-erased list of data representing the content of a

navigation stack.”

14. With appropriate example and code snippet, explain multi-column navigation with sidebar, content and detail.
15. How to navigate among multiple unrelated views? Explain with appropriate example and code snippet.